Lec 2 Linear Regression

Q1) B_1 Represent?

B)) The change in $\hat{y}$ for a one-unit increes in X

Q2) liner Regression is used to predict:

C) A continous numeric values

Q3) A residual is defined as:

B) $y - \hat{y}$

Q4) Linear regression chooses the line that minimizes:

A) The sum of squard residuals

Q5) $\bar{x} = 6 \quad \bar{y} = 65 \quad B_1 = 6,5 \quad B_0 \Rightarrow ?$

$B_0 = \bar{y} - B_1 \bar{x} = 65 - 6,5 \times 5 = 26$

B) 26

Q6) $\hat{y} = 26 + 6,5 x \quad x = 7$

$\hat{y} = 26 + 6,5 \times 7 = 71,5$

B) 71,5

Q7) If $\Sigma (y - \hat{y})^2 = 10$ over $n = 5$ the MSE is:

$MSE = \frac{\Sigma (y-\hat{y})^2}{n} = \frac{10}{5} = 2$

C) 2

Q8) An R^2 close to 1 means the model:

A) Explain almost all the variation in y.